Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The unit circle and radian measure

Convert 210° to radians.

2 [Foundational] • Target: The unit circle and radian measure

Convert $\frac{5\pi}{4}$ radians to degrees.

3 [Foundational] • Target: The unit circle and radian measure

Find the exact value of $\sin(240^\circ)$.

4 [Foundational] • Target: Trigonometric identities

If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos\theta$ and $\tan\theta$.

5 [Foundational] • Target: Double angle formulas

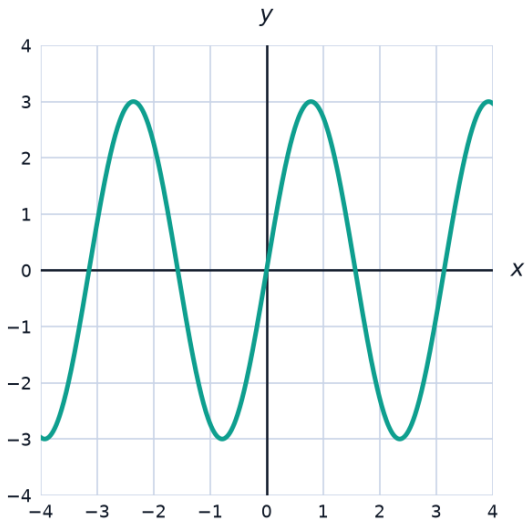
Given $\cos\theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

6 [Foundational] • Target: Double angle formulas

Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin\theta$ and $\cos\theta$ only.

7 [Foundational] • Target: Graphs of trigonometric functions

The graph shows $y = 3\sin(2x)$. State the amplitude and period.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Graphs of trigonometric functions

The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

9 [Intermediate] • Target: The Law of Sines

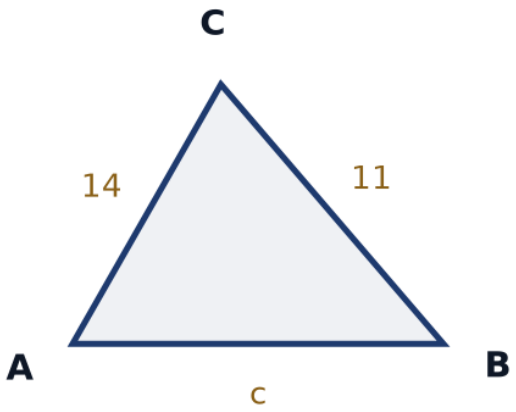
In triangle ABC, $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b, to the nearest hundredth.

10 [Intermediate] • Target: The Law of Sines

In triangle ABC, $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

11 [Intermediate] • Target: The Law of Cosines and the sine area formula

In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c, to the nearest hundredth, and the area of the triangle, to the nearest tenth.



12 [Intermediate] • Target: The Law of Cosines and the sine area formula

State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a, b, c and included angle C.

13 [Intermediate] • Target: Inverse trigonometric functions

Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.

14 [Intermediate] • Target: Inverse trigonometric functions

Find $\theta = \arctan(1)$ in degrees (principal value).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 6.01 Circles & Coordinate Geometry

This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.

16 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

In triangle ABC, $a = 8$, $\angle B = 35^\circ$, and $\angle C = 75^\circ$. Use the Law of Sines to calculate the length of side b to two decimal places.

17 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 8.01 Statistics & Probability

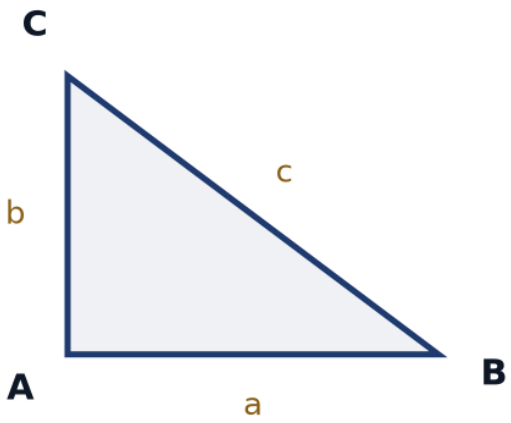
Two sides of a right triangle are 7 cm and 10 cm. If the third side is the hypotenuse, find: (a) the exact hypotenuse length, and (b) the measure of the smaller acute angle to the nearest tenth of a degree.

18 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 1.01 Advanced Algebra

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: The unit circle and radian measure • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



20 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A regular hexagon has side length $s = 6$ cm. (a) Calculate the exact area of the hexagon. (b) Find the length of the long diagonal connecting two opposite vertices.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The unit circle and radian measure

In triangle ABC, $a = 9$, $b = 15$, and $C = 110^\circ$. Find side c, to the nearest hundredth, and the area, to the nearest tenth.

a. $18.96, 61.4$

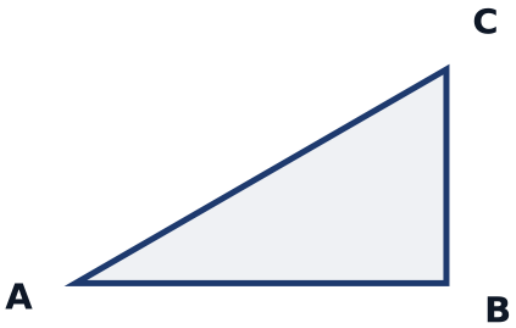
b. $19.96, 65.4$

c. $20.96, 63.4$

d. $19.96, 63.4$

22 [Foundational] • Target: Trigonometric identities

In ΔABC, $\angle A = 30^\circ$, $\angle B = 90^\circ$, and side $a = 6$. Find side b (Law of Sines).



a. 12

b. 13

c. 2

d. 3

23 [Foundational] • Target: Double angle formulas

In ΔABC, $\angle A = 90^\circ$, $a = 10$, and $b = 5$, with $\angle B$ acute. Find $\angle B$ (Law of Sines).

a. 118

b. 89

c. 53

d. 30

24 [Foundational] • Target: Solving trigonometric equations

In ΔABC, $a = 12$, $b = 9$, $c = 15$. Find $\angle C$ (Law of Cosines).

a. 47

b. 17

c. 90

d. 18

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25 [Foundational] • Target: Graphs of trigonometric functions

In ΔABC, $a = 8$, $b = 5$, and $\angle C = 60^\circ$. Find side c (Law of Cosines).

a. 1

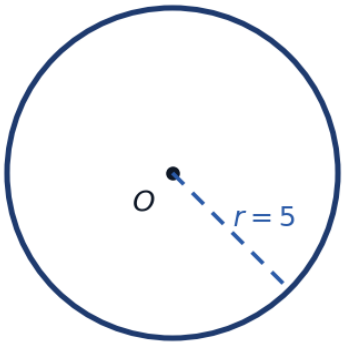
b. 19

c. 7

d. 3

26 [Foundational] • Target: The Law of Sines

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

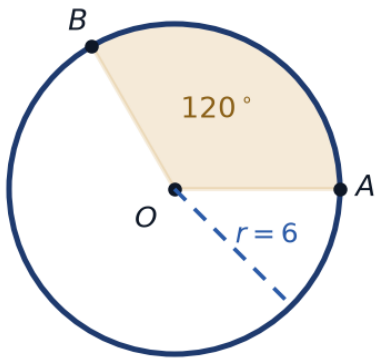
c. 1

d. $\frac{\sqrt{2}}{2}$

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27 [Foundational] • Target: The Law of Cosines and the sine area formula

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?



a. $\cos\theta$

b. $\sin\theta$

c. $\tan\theta$

d. 1

28 [Foundational] • Target: Inverse trigonometric functions

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15^\circ$.

a. -1

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: locating a ship from two lighthouses

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

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30 [Intermediate] • Target: The unit circle and radian measure

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

31 [Intermediate] • Target: Trigonometric identities

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$

d. $\frac{4}{5}$

32 [Intermediate] • Target: Double angle formulas

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

33 [Intermediate] • Target: Solving trigonometric equations

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

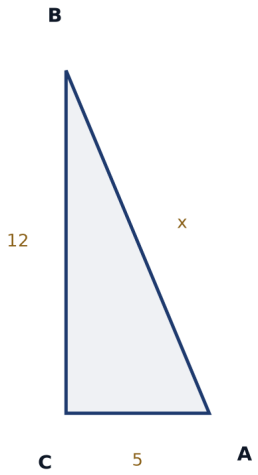
c. 1

d. 5/3

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34 [Intermediate] • Target: Graphs of trigonometric functions

In the diagram below, triangle ABC is a right-angled triangle with the right angle at C. Given that AC = 5, BC = 12, and angle BAC = θ, find the value of $\sin(\theta)$.



a. $\frac{12}{13}$

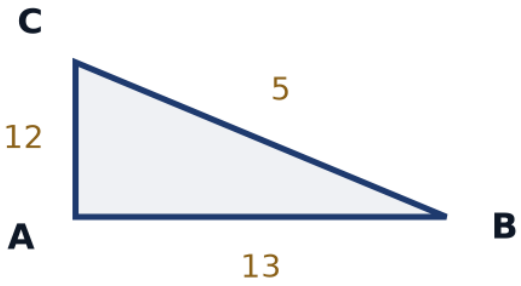
b. $\frac{5}{13}$

c. $\frac{12}{5}$

d. $\frac{5}{12}$

35 [Intermediate] • Target: The Law of Sines

In the right-angled triangle ABC shown below, the length of the side adjacent to angle A is 12 units and the length of the hypotenuse is 13 units. What is the cosine of angle A?



a. $\frac{5}{13}$

b. $\frac{12}{13}$

c. $\frac{5}{12}$

d. $\frac{13}{5}$

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36 [Intermediate] • Target: The Law of Cosines and the sine area formula

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

37 [Intermediate] • Target: Inverse trigonometric functions

State the double-angle formula $\sin 2\theta =$

a. $2\cos^2\theta - 1$

b. $\sin^2\theta - \cos^2\theta$

c. $2\sin\theta\cos\theta$

d. $\cos^2\theta - \sin^2\theta$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: locating a ship from two lighthouses • Cross-Lesson: 1.01 Advanced Algebra

In the diagram, ΔABC is a right-angled triangle with the right angle at C. The length of the hypotenuse AB is 26 units. Point D lies on AB such that CD is the altitude to the hypotenuse. If $AD = 4$ units and $BD = x$ units, find the value of x.

a. 12

b. 13

c. 16

d. 22

39 [Challenging] • Target: The unit circle and radian measure • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

In the diagram, two chords AB and CD intersect at point P inside a circle. If $PA = 6$, $PB = 8$, and $PC = 4$, what is the length of PD?

a. 6

b. 8

c. 12

d. 10

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40 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Consider a circle with center O. Two chords AB and CD intersect at point P inside the circle. If AP = 3, PB = 8, and CP = 4, determine the length of PD. The diagram shows a circle with intersecting chords.

a. 6

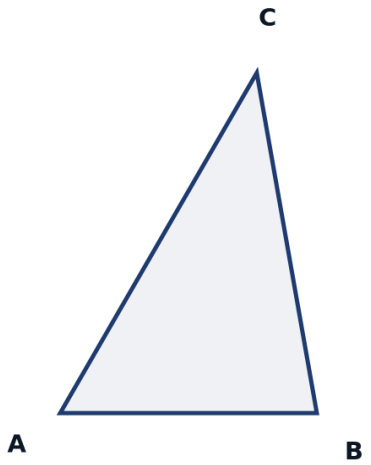
b. 8

c. 10

d. 12

41 [Challenging] • Target: Double angle formulas • Cross-Lesson: 4.01 Lines, Triangles & Polygons

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.



a. 5.7

b. 6.7

c. 7.7

d. 4.7

42 [Challenging] • Target: Solving trigonometric equations • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Two similar solids have a surface-area ratio of $25:9$. Find their volume ratio.

a. 125:1

b. 125:27

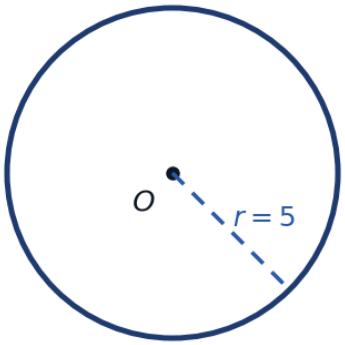
c. 125:729

d. 125:216

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43 [Challenging] • Target: Graphs of trigonometric functions • Cross-Lesson: 7.01 Solid Geometry & Similarity

In the diagram below, a circle has its center at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the radius of the circle?



a. 5

b. 7

c. 25

d. 12

44 [Challenging] • Target: The Law of Sines • Cross-Lesson: 8.01 Statistics & Probability

In a right-angled triangle, one of the acute angles is 30°. What is the measure of the other acute angle?

a. 30°

b. 45°

c. 60°

d. 90°

45 [Challenging] • Target: The Law of Cosines and the sine area formula • Cross-Lesson: 1.01 Advanced Algebra

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓

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